

Solving Large-Scale Electricity Market Pricing Problems in Polynomial Time

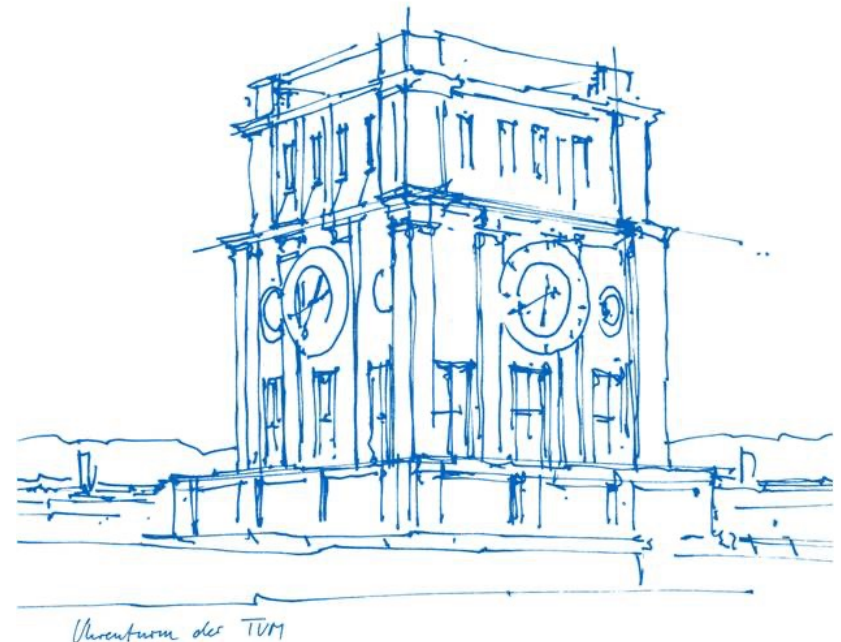
Mete Ahunbay, Martin Bichler, Teodora Dobos, Johannes Knörr

TUM School of Computation, Information and Technology

Chair of Decision Sciences & Systems

OR 2024

September 2024



Pricing in non-convex markets

- Electricity markets are **non-convex markets**:
 - non-convex preferences: minimum production levels, startup-costs, etc.
- **Day-ahead markets**: compute an allocation and prices for the next day

Two challenges:



I. Identifying an **optimal** allocation

- Scalability issues

Yes, with the ***markup mechanism!***



II. Identifying **equilibria (market-clearing prices)**

- Desirable properties

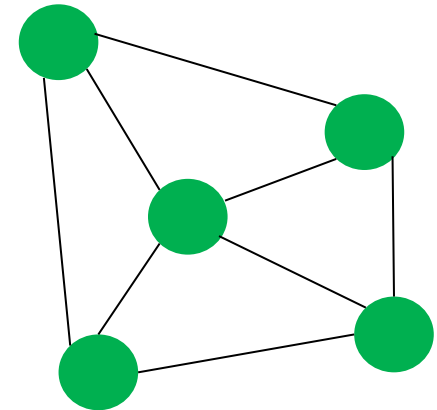
Can we efficiently find a near-optimal allocation and prices that satisfy budget balance?

Outline

1. **Electricity market model**
2. Welfare maximization & pricing problems
3. Markup mechanism
4. Evaluation
5. Conclusions

Electricity market model

- Transmission network
 - Periods (24 hours)
- } Goods: electricity at each node for every period

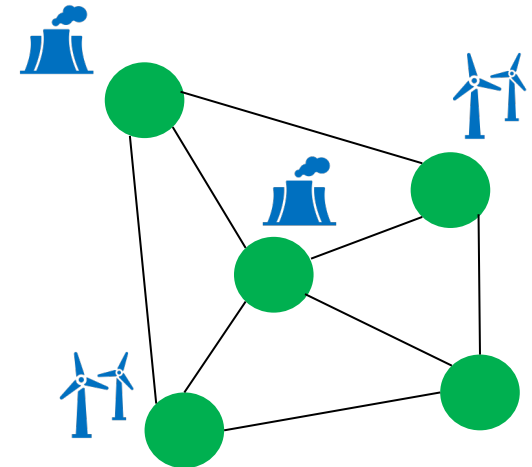
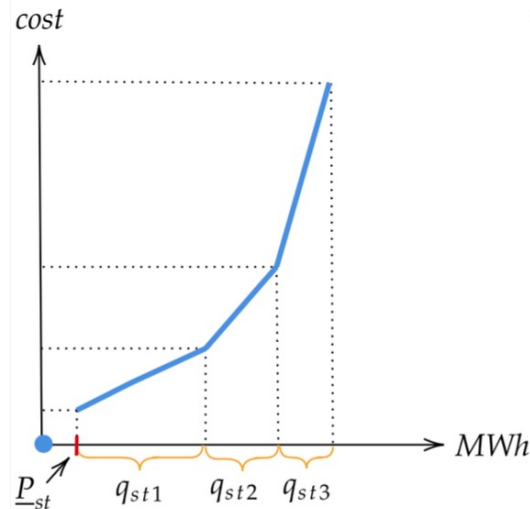


Electricity market model

- Transmission network
 - Periods (24 hours)
- } Goods: electricity at each node for every period

1. Set of sellers S

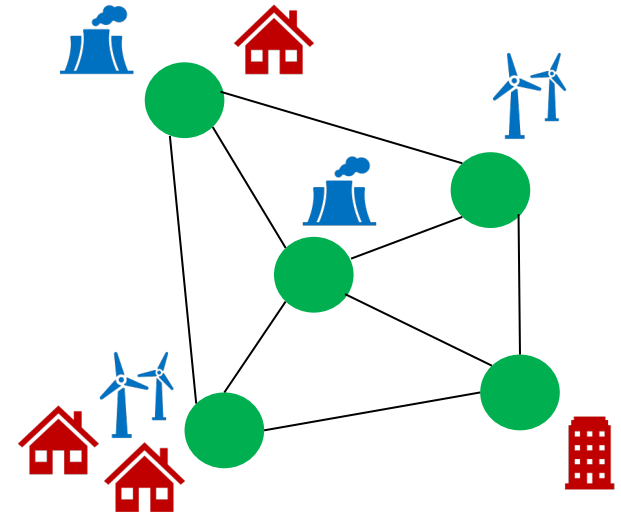
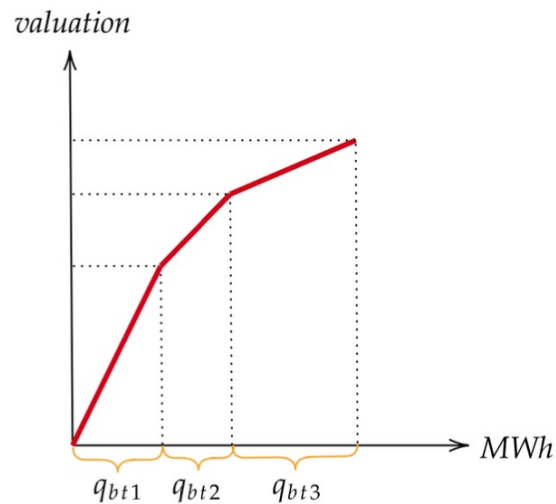
- production constraints (min-uptime, no-load cost, etc.)
- supply bids $\rightarrow$ cost function



Electricity market model

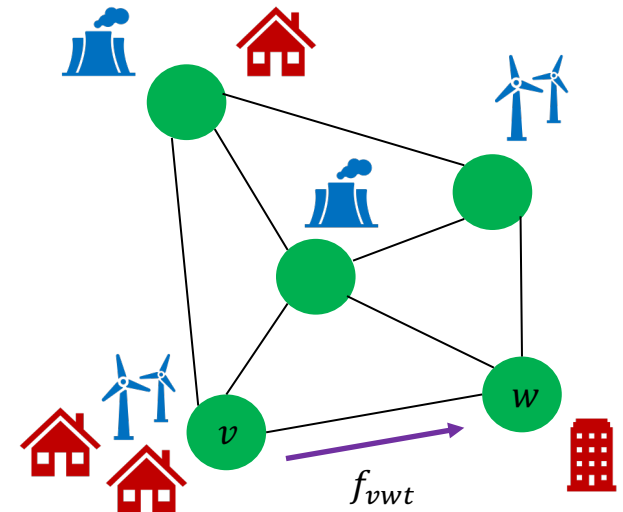
- Transmission network
 - Periods (24 hours)
- } Goods: electricity at each node for every period

1. Set of sellers S
2. Set of buyers B
 - price-inelastic demand
 - price-elastic demand bids $\rightarrow$ valuation function



Electricity market model

- Transmission network
 - Periods (24 hours)
- } Goods: electricity at each node for every period
1. Set of sellers S
 2. Set of buyers B
 3. DCOPF power flow approximation (Molzahn & Hiskens, 2019)
 - linear approximation



Outline

1. Electricity market model
- 2. Welfare maximization & pricing problems**
3. Markup mechanism
4. Evaluation
5. Conclusions

Welfare maximization problem (WMP)

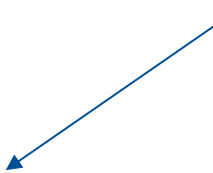
Let $z = (x, y, u, f)$ be an allocation with

- x : buyers' fulfilled demand
- y : sellers' production
- u : sellers' commitment
- f : flows

WMP is defined as

$$\begin{array}{ll} \max_z & W(z) \\ \text{s. t.} & z \in \Omega \\ & u_{st} \in \{0, 1\} \forall s \in S, t \in T \end{array}$$

Node-balance constraints
Price-inelastic demand fulfillment
Production constraints, etc.



where the welfare is

$$W(z) = \sum_{b \in B} v_b(x_b) - \sum_{s \in S} c_s(y_s)$$

Pricing problem

- For a **convex market**, the *Welfare Theorems* (Arrow & Debreu 1954) hold.
- A primal solution z^* and a dual solution p^* form a **Walrasian equilibrium**.
- Such a pair (z^*, p^*) satisfies certain desirable properties:



Efficiency



Supply-demand
balance



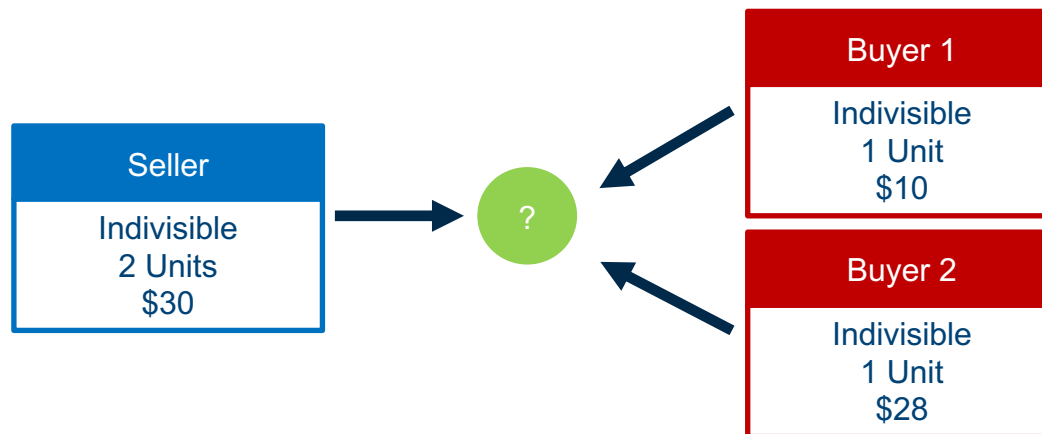
Envy-freeness



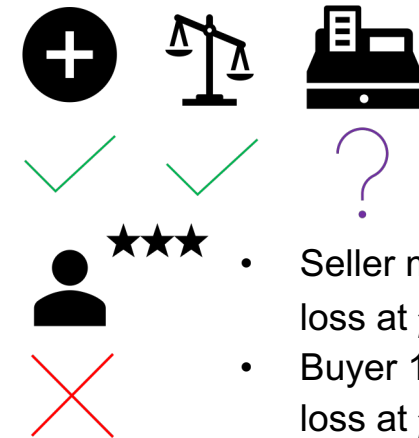
Budget balance

Walrasian equilibria in electricity markets?

With *non-convexities* present, a Walrasian equilibrium might not exist.



Clearing the market



- Seller makes a loss at $p \leq \$10$
- Buyer 1 makes a loss at $p > \$10$

Individual rationality in electricity markets

- No guarantee that prices satisfy **individual rationality** $\Leftrightarrow$ market participants make no loss
- **Make-whole payments (MWPs)**: compensations for the negative utility of a market participant ℓ , given prices p and allocation z_ℓ^* :

$$MWP_\ell(p|z_\ell^*) = \max \{-u_\ell(z_\ell^*|p), 0\}$$

Individual rationality in electricity markets

- No guarantee that prices satisfy **individual rationality** $\Leftrightarrow$ market participants make no loss
- **Make-whole payments (MWPs)**: compensations for the negative utility of a market participant ℓ , given prices p and allocation z_ℓ^* :

$$MWP_\ell(p|z_\ell^*) = \max \{-u_\ell(z_\ell^*|p), 0\}$$

- Budget balance: market requires no external money to pay MWPs

How to compute prices?

Integer Programming (IP) Pricing (O'Neill et al., 2005)

- standard approach used in the US
- “marginal pricing” for non-convex markets
- good congestion signals, but large MWPs

Outline

1. Electricity market model
2. Welfare maximization & pricing problems
- 3. Markup mechanism**
4. Evaluation
5. Conclusions

Markup mechanism

Observation: If no Walrasian equilibrium exists, any (price, allocation) pair must feature either:

- $\text{supply} > \text{demand} \Rightarrow \text{budget deficit}$
- $\text{demand} > \text{supply} \Rightarrow \text{rationing}$

Milgrom & Watt, 2022 proposed the [markup mechanism](#) for [two-sided uncoupled markets](#) – characteristics:

- near-optimal allocation
- buyers pay higher prices to cover the excess supply
- convex optimization

Markup mechanism for electricity markets

Phase 1: *Valuations scaling & convexifying WMP*

$$\begin{aligned} \max_z \quad & W(z, \alpha) \\ \text{s. t.} \quad & z \in \Omega \\ & 0 \leq u_s \leq 1 \quad \forall s \in S \end{aligned}$$

$$W(z, \alpha) = \sum_{b \in B} \frac{v_b(x_b)}{1 + \alpha} - \sum_{s \in S} c_s(y_s)$$

Phase 2: *Rounding & Residual Clearing*

- *Option 1: Rounding with a threshold $\delta \in (0, 1]$*
- *Option 2: Rounding by solving a small MILP*

Markup mechanism for electricity markets

Phase 1: Valuations scaling & convexifying WMP

$$\begin{array}{ll} \max_z & W(z, \alpha) \\ \text{s. t.} & z \in \Omega \\ & 0 \leq u_s \leq 1 \quad \forall s \in S \end{array}$$

→ obtain $(u'_s)_{s \in S}$

Phase 2: Rounding & Residual Clearing – *Option 1: Rounding with a threshold $\delta \in (0, 1]$*

$$\begin{array}{ll} \max_z & W(z) \\ \text{s. t.} & z \in \Omega \\ & u_s = u_s^\alpha \quad \forall s \in S \end{array}$$

$$u_s^\alpha = 1 \text{ if } u'_s > \delta, \text{ else } u_s^\alpha = 0$$


Markup mechanism for electricity markets

Phase 1: Valuations scaling & convexifying WMP

$$\begin{aligned} \max_z \quad & W(z, \alpha) \\ \text{s. t.} \quad & z \in \Omega \\ & 0 \leq u_s \leq 1 \quad \forall s \in S \end{aligned}$$

→ obtain $(u'_s)_{s \in S}$

Phase 2: Rounding & Residual Clearing – *Option 2: Rounding by solving a small MILP*

$$\begin{aligned} \max_z \quad & W(z) \\ \text{s. t.} \quad & z \in \Omega \\ & u_s = u_s^\alpha \quad \forall s \in S^\alpha \\ & u_s \in \{0, 1\} \quad \forall s \in S \setminus S^\alpha \end{aligned}$$

$S^\alpha = \{s \in S \mid u'_s \in \{0, 1\}\}$



What about the prices?

- Prices are obtained from the dual problem of the convexified WMP.

$$\begin{aligned} \max_z \quad & W(z, \alpha) \\ \text{s. t.} \quad & z \in \Omega \\ & 0 \leq u_s \leq 1 \quad \forall s \in S \end{aligned}$$

- Sellers' prices are p and buyers' prices are $(1 + \alpha)p$.
- Role of α : ensure that the market has no budget deficit:

$$p^T \left(\sum_b x_b (1 + \alpha) - \sum_y y_s \right) - MWPs \geq 0$$

→ No guarantee that such α exists!

Outline

1. Electricity market model
2. Welfare maximization & pricing problems
3. Markup mechanism
- 4. Evaluation**
5. Conclusions

Data sets

6 scenarios from the ARPA-E Challenge 2 extended to 24 hours

Scenario	Nodes	Lines	Sellers	Buyers	Demand (GWh)	Supply (GWh)
1.	617	841	94	404	506	830
2.	2,020	2,709	194	1,392	622	1,041
3.	8,718	26,084	321	5,819	988	1,088
4.	10,480	16,253	750	6,871	2,033	2,289
5.	12,209	30,500	2,009	6,987	2,391	3,328
6.	18,889	57,630	898	12,273	2,686	2,916

Data sets

6 scenarios from the ARPA-E Challenge 2 extended to 24 hours



Scenario	Nodes	Lines	Sellers	Buyers	Demand (GWh)	Supply (GWh)
1.	617	841	94	404	506	830
2.	2,020	2,709	194	1,392	622	1,041
3.	8,718	26,084	321	5,819	988	1,088
4.	10,480	16,253	750	6,871	2,033	2,289
5.	12,209	30,500	2,009	6,987	2,391	3,328
6.	18,889	57,630	898	12,273	2,686	2,916
ISO-NE	1,216	-	98	139	380	2,151
PJM	13,431	-	1,217	366	2,204	4,907

Experiments overview

Algorithm	S=D	Welfare	Supply	Oversupply	MWPs	Payments sellers	Budget deficit	α	δ	RWL	Time (s)	MIP Gap
617 nodes												
OPT	y	238,195,146.54	298,663.25	0	86,336.04	12,197,605.05	86,336.04	–	–	–	138.06	9.90E–05
RC- δ	y	238,174,692.20	299,450.34	0	47,262.63	12,266,027.18	47,262.63	0	0.01	0.01%	2.64	–
RC- δ	y	238,104,566.44	289,878.27	0	4,519.08	11,859,915.41	(190,312.38)	0.01	0.5	0.04%	4.75	–
RC- δ	y	238,075,360.52	284,765.32	0	4,914.59	11,455,385.53	(1,781,451.71)	0.1	0.5	0.04%	2.45	–
RC-MILP	y	238,174,620.47	299,264.85	0	47,966.66	12,247,568.41	(7,853,257.48)	0.01	–	0.01%	10.08	2.18E–04
2,020 nodes												
OPT	y	597,582,514.99	284,313.42	0	149,029.65	5,768,803.50	149,029.65	–	–	–	795.28	8.92E–05
RC- δ	y			infeasible phase 2 for $\alpha \in \{0.01, 0.1\}$					0.5			–
RC- δ	y	597,567,977.75	285,761.50	0	71,262.31	5,740,925.92	(34,085.29)	0.01	0.1	0.00%	11.09	–
RC- δ	y	597,575,055.59	284,209.02	0	66,401.13	5,632,556.47	(957,183.18)	0.1	0.1	0.00%	23.13	–
RC-MILP	y	597,566,188.64	285,806.59	0	71,917.06	5,742,909.07	(4,825,879.65)	0.01	–	0.00%	57.25	1.19E–04
8,718 nodes												
OPT	y	53,762,905.20	803,462.34	0	11,851.68	27,947,069.37	11,851.68	–	–	–	3,108.48	2.45E–05
RC- δ	y	53,762,321.96	803,165.67	0	48.61	27,836,077.35	(310,681.06)	0.01	0.2, 0.5	0.00%	333.49	–
RC-MILP	y	53,757,420.03	800,556.51	0	40.77	26,038,052.89	(5,749,599.77)	0.1	–	0.1%	1,244.35	2.84E–05
10,480 nodes												
OPT	y	82,630,373.56	1,610,608.60	0	56,396.69	45,419,264.48	56,396.69				14,454.78	5.89E–04
RC- δ	y	82,628,972.82	1,609,101.04	0	5,089.26	44,732,094.94	(512,300.75)	0.01	0.5	0.00%	2,271.02	–
RC-MILP	y	no incumbent solution found within 1h (phase 2)						0.01	–			
12,209 nodes												
OPT	y	N/A										
RC- δ	y	901,291,505.82	1,227,340.33	0	237,410.14	36,245,838.24	(352,738.44)	0.01	0.01	N/A	1,445.82	–
RC-MILP	y	901,294,362.51	1,227,254.09	0	233,892.81	36,240,638.93	(23,124,930.62)	0.01	–	N/A	6,798.37	1.95E–04
18,889 nodes												
OPT	y	N/A										
RC- δ	y	54,586,238.75	2,081,555.98	0	20,477.98	35,938,241.61	(377,467.48)	0.01	0.01	N/A	49,178.68	–

Result 1: Close approximation of OPT

Relative welfare loss

Algorithm	S=D	Welfare	Supply	Oversupply	MWPs	Payments sellers	Budget deficit	α	δ		RWL	Time (s)	MIP Gap	
617 nodes														
OPT	y	238,195,146.54	298,663.25	0	86,336.04	12,197,605.05	86,336.04	–	–		–	138.06	9.90E–05	
RC- δ	y	238,174,692.20	299,450.34	0	47,262.63	12,266,027.18	47,262.63	0	0.01		0.01%	2.64	–	
RC- δ	y	238,104,566.44	289,878.27	0	4,519.08	11,859,915.41	(190,312.38)	0.01	0.5		0.04%	4.75	–	
RC- δ	y	238,075,360.52	284,765.32	0	4,914.59	11,455,385.53	(1,781,451.71)	0.1	0.5		0.04%	2.45	–	
RC-MILP	y	238,174,620.47	299,264.85	0	47,966.66	12,247,568.41	(7,853,257.48)	0.01	–		0.01%	10.08	2.18E–04	
2,020 nodes														
OPT	y	597,582,514.99	284,313.42	0	149,029.65	5,768,803.50	149,029.65	–	–		–	795.28	8.92E–05	
RC- δ	y			infeasible phase 2 for $\alpha \in \{0.01, 0.1\}$						0.5			–	
RC- δ	y	597,567,977.75	285,761.50	0	71,262.31	5,740,925.92	(34,085.29)	0.01	0.1		0.00%	11.09	–	
RC- δ	y	597,575,055.59	284,209.02	0	66,401.13	5,632,556.47	(957,183.18)	0.1	0.1		0.00%	23.13	–	
RC-MILP	y	597,566,188.64	285,806.59	0	71,917.06	5,742,909.07	(4,825,879.65)	0.01	–		0.00%	57.25	1.19E–04	
8,718 nodes														
OPT	y	53,762,905.20	803,462.34	0	11,851.68	27,947,069.37	11,851.68	–	–		–	3,108.48	2.45E–05	
RC- δ	y	53,762,321.96	803,165.67	0	48.61	27,836,077.35	(310,681.06)	0.01	0.2, 0.5		0.00%	333.49	–	
RC-MILP	y	53,757,420.03	800,556.51	0	40.77	26,038,052.89	(5,749,599.77)	0.1	–		0.1%	1,244.35	2.84E–05	
10,480 nodes														
OPT	y	82,630,373.56	1,610,608.60	0	56,396.69	45,419,264.48	56,396.69					14,454.78	5.89E–04	
RC- δ	y	82,628,972.82	1,609,101.04	0	5,089.26	44,732,094.94	(512,300.75)	0.01	0.5		0.00%	2,271.02	–	
RC-MILP	y	no incumbent solution found within 1h (phase 2)							0.01	–				
12,209 nodes														
OPT	y	N/A												
RC- δ	y	901,291,505.82	1,227,340.33	0	237,410.14	36,245,838.24	(352,738.44)	0.01	0.01		N/A	1,445.82	–	
RC-MILP	y	901,294,362.51	1,227,254.09	0	233,892.81	36,240,638.93	(23,124,930.62)	0.01	–		N/A	6,798.37	1.95E–04	
18,889 nodes														
OPT	y	N/A												
RC- δ	y	54,586,238.75	2,081,555.98	0	20,477.98	35,938,241.61	(377,467.48)	0.01	0.01		N/A	49,178.68	–	

Result 2: Fast method

Algorithm	S=D	Welfare	Supply	Oversupply	MWPs	Payments sellers	Budget deficit	α	δ	RWL	Time (s)	MIP Gap
617 nodes												
OPT	y	238,195,146.54	298,663.25	0	86,336.04	12,197,605.05	86,336.04	–	–	–	138.06	9.90E–05
RC- δ	y	238,174,692.20	299,450.34	0	47,262.63	12,266,027.18	47,262.63	0	0.01	0.01%	2.64	–
RC- δ	y	238,104,566.44	289,878.27	0	4,519.08	11,859,915.41	(190,312.38)	0.01	0.5	0.04%	4.75	–
RC- δ	y	238,075,360.52	284,765.32	0	4,914.59	11,455,385.53	(1,781,451.71)	0.1	0.5	0.04%	2.45	–
RC-MILP	y	238,174,620.47	299,264.85	0	47,966.66	12,247,568.41	(7,853,257.48)	0.01	–	0.01%	10.08	2.18E–04
2,020 nodes												
OPT	y	597,582,514.99	284,313.42	0	149,029.65	5,768,803.50	149,029.65	–	–	–	795.28	8.92E–05
RC- δ	y			infeasible phase 2 for $\alpha \in \{0.01, 0.1\}$						0.5		–
RC- δ	y	597,567,977.75	285,761.50	0	71,262.31	5,740,925.92	(34,085.29)	0.01	0.1	0.00%	11.09	–
RC- δ	y	597,575,055.59	284,209.02	0	66,401.13	5,632,556.47	(957,183.18)	0.1	0.1	0.00%	23.13	–
RC-MILP	y	597,566,188.64	285,806.59	0	71,917.06	5,742,909.07	(4,825,879.65)	0.01	–	0.00%	57.25	1.19E–04
8,718 nodes												
OPT	y	53,762,905.20	803,462.34	0	11,851.68	27,947,069.37	11,851.68	–	–	–	3,108.48	2.45E–05
RC- δ	y	53,762,321.96	803,165.67	0	48.61	27,836,077.35	(310,681.06)	0.01	0.2, 0.5	0.00%	333.49	–
RC-MILP	y	53,757,420.03	800,556.51	0	40.77	26,038,052.89	(5,749,599.77)	0.1	–	0.1%	1,244.35	2.84E–05
10,480 nodes												
OPT	y	82,630,373.56	1,610,608.60	0	56,396.69	45,419,264.48	56,396.69				14,454.78	5.89E–04
RC- δ	y	82,628,972.82	1,609,101.04	0	5,089.26	44,732,094.94	(512,300.75)	0.01	0.5	0.00%	2,271.02	–
RC-MILP	y	no incumbent solution found within 1h (phase 2)							0.01	–		
12,209 nodes												
OPT	y	N/A										
RC- δ	y	901,291,505.82	1,227,340.33	0	237,410.14	36,245,838.24	(352,738.44)	0.01	0.01	N/A	1,445.82	–
RC-MILP	y	901,294,362.51	1,227,254.09	0	233,892.81	36,240,638.93	(23,124,930.62)	0.01	–	N/A	6,798.37	1.95E–04
18,889 nodes												
OPT	y	N/A										
RC- δ	y	54,586,238.75	2,081,555.98	0	20,477.98	35,938,241.61	(377,467.48)	0.01	0.01	N/A	49,178.68	–

Result 3: No budget deficit

*A quantity in paratheses indicates budget surplus

Algorithm	S=D	Welfare	Supply	Oversupply	MWPs	Payments sellers	Budget deficit	α	δ	RWL	Time (s)	MIP Gap
617 nodes												
OPT	y	238,195,146.54	298,663.25	0	86,336.04	12,197,605.05	86,336.04	–	–	–	138.06	9.90E−05
RC- δ	y	238,174,692.20	299,450.34	0	47,262.63	12,266,027.18	47,262.63	0	0.01	0.01%	2.64	–
RC- δ	y	238,104,566.44	289,878.27	0	4,519.08	11,859,915.41	(190,312.38)	0.01	0.5	0.04%	4.75	–
RC- δ	y	238,075,360.52	284,765.32	0	4,914.59	11,455,385.53	(1,781,451.71)	0.1	0.5	0.04%	2.45	–
RC-MILP	y	238,174,620.47	299,264.85	0	47,966.66	12,247,568.41	(7,853,257.48)	0.01	–	0.01%	10.08	2.18E−04
2,020 nodes												
OPT	y	597,582,514.99	284,313.42	0	149,029.65	5,768,803.50	149,029.65	–	–	–	795.28	8.92E−05
RC- δ	y			infeasible phase 2 for $\alpha \in \{0.01, 0.1\}$						0.5		–
RC- δ	y	597,567,977.75	285,761.50	0	71,262.31	5,740,925.92	(34,085.29)	0.01	0.1	0.00%	11.09	–
RC- δ	y	597,575,055.59	284,209.02	0	66,401.13	5,632,556.47	(957,183.18)	0.1	0.1	0.00%	23.13	–
RC-MILP	y	597,566,188.64	285,806.59	0	71,917.06	5,742,909.07	(4,825,879.65)	0.01	–	0.00%	57.25	1.19E−04
8,718 nodes												
OPT	y	53,762,905.20	803,462.34	0	11,851.68	27,947,069.37	11,851.68	–	–	–	3,108.48	2.45E−05
RC- δ	y	53,762,321.96	803,165.67	0	48.61	27,836,077.35	(310,681.06)	0.01	0.2, 0.5	0.00%	333.49	–
RC-MILP	y	53,757,420.03	800,556.51	0	40.77	26,038,052.89	(5,749,599.77)	0.1	–	0.1%	1,244.35	2.84E−05
10,480 nodes												
OPT	y	82,630,373.56	1,610,608.60	0	56,396.69	45,419,264.48	56,396.69				14,454.78	5.89E−04
RC- δ	y	82,628,972.82	1,609,101.04	0	5,089.26	44,732,094.94	(512,300.75)	0.01	0.5	0.00%	2,271.02	–
RC-MILP	y	no incumbent solution found within 1h (phase 2)						0.01	–			
12,209 nodes												
OPT	y	N/A										
RC- δ	y	901,291,505.82	1,227,340.33	0	237,410.14	36,245,838.24	(352,738.44)	0.01	0.01	N/A	1,445.82	–
RC-MILP	y	901,294,362.51	1,227,254.09	0	233,892.81	36,240,638.93	(23,124,930.62)	0.01	–	N/A	6,798.37	1.95E−04
18,889 nodes												
OPT	y	N/A										
RC- δ	y	54,586,238.75	2,081,555.98	0	20,477.98	35,938,241.61	(377,467.48)	0.01	0.01	N/A	49,178.68	–

Outline

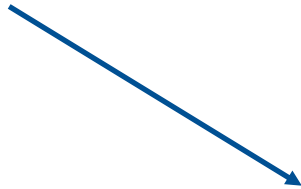
1. Electricity market model
2. Welfare maximization & pricing problems
3. Markup mechanism
4. Evaluation
- 5. Conclusions**

Conclusions

- Electricity markets need to solve very large non-convex problems and determine prices.
- Runtime limits are a concern.
- The markup mechanism computes a near-optimal allocation in polynomial time.
- The markup mechanism relies on two hyperparameters.
- Buyers pay slightly higher prices to cover the make-whole payments.
- The markup mechanism is a practical recommendation for large electricity markets.

Thanks!

Our paper



References

- Milgrom P, Watt M (2022) Walrasian mechanisms for non-convex economies and the bound-form first welfare theorem. *Proceedings of the ACM Conference of Economics and Computation*
- Neill, Hytowitz, Whitman, Mead, Dautel, Chen, Eldridge, Siskind, Kheloussi, Kolkman, Smith, Castillo, Mays: Essays on average incremental cost pricing for independent system operators. 2019
- Ahunbay, Bichler, Knoerr: Pricing optimal outcomes in coupled and non-convex markets: Theory and applications to electricity markets, *Operations Research* 2024, <https://doi.org/10.1287/opre.2023.0401>
- Molzahn, D.K., Hiskens, I.A., 2019. A survey of relaxations and approximations of the power flow equations. *Found. Trends® Electr. Energy Syst.* 4 (1–2), 1–221. <http://dx.doi.org/10.1561/31000000012>.
- Liberopoulos, Andrianesis: Critical review of pricing schemes in markets with non-convex costs. 2016. *Operations Research* 64(1):17–31
- Starr RM, Quasi-equilibria in markets with non-convex preferences. *Econometrica* 37(1):25, ISSN 00129682, URL <http://dx.doi.org/10.2307/1909201>.
- O'Neill RP, Sotkiewicz PM, Hobbs BF, Rothkopf MH, Stewart WR (2005) Efficient market-clearing prices in markets with nonconvexities. *European Journal of Operational Research* 1(164):269–285, URL <https://doi.org/10.1016/j.ejor.2003.12.011>